
INVENTION DISCLOSURE DOCUMENT

SafePoint

A union-owned real-time workplace safety incident reporting and bargaining-intelligence platform for K–12 education.

INVENTORS	Sonny Steele · Nick Sireci
DATE OF DISCLOSURE	May 29, 2026
DOCUMENT STATUS	Confidential — for IP counsel review
VERSION	1.0
CLASSIFICATION	Software invention — system, method, data architecture

NOTICE. This document is the inventors' contemporaneous, dated record of an invention, prepared for the purpose of obtaining professional intellectual-property counsel. It is not itself a patent, trademark registration, or copyright filing. The contents are confidential and proprietary to the inventors. Disclosure to third parties without a written non-disclosure agreement may impair patent rights.

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1. Document purpose and disclaimers

This Invention Disclosure Document (“IDD”) records the conception and reduction to practice of the software system identified herein as **SafePoint**. Its purpose is to establish a contemporaneous, dated, signed record of the invention sufficient to support subsequent intellectual-property filings, including but not limited to United States provisional and non-provisional patent applications, trademark registrations, and copyright registrations.

What this document is

An evidentiary record describing the invention with sufficient specificity to (i) identify what is novel, (ii) distinguish the invention from prior art, (iii) describe at least one workable embodiment, and (iv) support the prosecution of patent, trademark, and copyright applications by qualified counsel.

What this document is not

This document is **not** a patent application, a patent grant, a trademark registration, a copyright registration, or legal advice. It does not, by itself, confer any enforceable intellectual-property right. Neither inventor is a licensed patent attorney. Filings derived from this disclosure should be drafted and prosecuted by a registered patent attorney or agent admitted to practice before the United States Patent and Trademark Office (USPTO).

Confidentiality

This document and the information it contains are confidential and proprietary to the inventors. Public disclosure of the invention—through publication, commercial offer for sale, or public use—starts a one-year statutory bar in the United States under 35 U.S.C. § 102(b) and may immediately bar patent rights in most other jurisdictions. Disclosure to any third party should be preceded by a written non-disclosure agreement.

2. Inventor information

Joint inventors	Sonny Steele and Nick Sireci
Mailing address (Sonny Steele)	[To be completed]
Email (Sonny Steele)	[To be completed]
Citizenship (Sonny Steele)	[To be completed]
Mailing address (Nick Sireci)	[To be completed]
Email (Nick Sireci)	[To be completed]
Citizenship (Nick Sireci)	[To be completed]
Employer(s) at conception	[To be completed — consult each employment agreement re: assignment]
Date of conception	[Earliest date the invention was conceived — to be completed]
Date of first reduction to practice	Approximately May 2026 (working software prototype)
Date of first written disclosure	May 29, 2026

Action required: Complete every '[To be completed]' field above before executing the signature page in Section 15. Each joint inventor's contribution should be separately documented — see the contribution statement below — and reviewed with counsel before filing. Misnaming or omitting an inventor is grounds for invalidating a granted patent under 35 U.S.C. § 256; conversely, naming a non-inventor as an inventor is also a defect curable only by correction.

Contribution statement

Joint inventorship under 35 U.S.C. § 116 does not require equal contribution; it requires that each named inventor contributed to the conception of at least one claim. Each joint inventor should briefly describe their contribution below; counsel will rely on these statements when mapping inventors to specific claims in any future filing.

Sonny Steele — contribution	[To be completed by Sonny Steele — describe the specific concepts, architectural decisions, or implementation work contributed.]
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Nick Sireci — contribution

[To be completed by Nick Sireci — describe the specific concepts, architectural decisions, or implementation work contributed.]

3. Title and field of invention

Working title

SafePoint

Long-form descriptive title

“System and Method for Union-Owned Real-Time Workplace Safety Incident Reporting, Multi-Role Notification Fan-Out, and Bargaining-Intelligence Analytics in K–12 Educational Settings.”

Field of invention

The invention relates to enterprise software, specifically to incident-management systems for workplace safety in education; to data architectures that segregate labor-organization-owned data from employer-owned data; to multi-role real-time notification systems with role- and importance-gated message routing; to compliance analytics under FERPA, HIPAA, and OSHA frameworks; and to causation-attribution analytics that combine first-party incident telemetry with third-party epidemiological research.

USPC / CPC indicators (advisory)

Counsel may consider the following classification areas as starting points: G06Q 10/06 (Resources, workflows, human or project management); G06Q 10/107 (Computer-aided management of electronic messages); G06Q 50/20 (Education); G06Q 50/22 (Healthcare). Final classification is the examiner's prerogative.

4. Background and state of the art

Workplace violence and safety incidents against K–12 educators have reached levels that public-health researchers describe as a national crisis. The American Psychological Association's 2024 Task Force on Violence Against Educators reported that 80% of teachers experienced verbal or threatening aggression and 56% experienced physical violence during the 2021–22 academic year. The National Center for Education Statistics reported that the rate of student verbal and threatening behavior toward classroom teachers nearly doubled between 2009–10 and 2019–20. Recent peer-reviewed research (Reddy et al., 2024) found that 49.5% of paraprofessionals had experienced physical violence at work.

Despite the magnitude of the problem, the systems used to record these incidents are almost universally **owned, operated, and controlled by the employer** (the school district). District-side incident-management platforms such as Frontline Education's suite, Origami Risk, SafetyCulture, RiskCloud, and various proprietary district-developed forms collect incident data and present it back to district administrators. The same employer whose practices may be the subject of the incident report controls the dashboards, retention, access policies, and analytical lens applied to the data.

Labor organizations representing the affected workforce—teachers' unions and education associations such as the NEA, AFT, and their state and local affiliates—have no equivalent system. When a union wishes to prepare a bargaining brief, file an OSHA General Duty Clause complaint, or support a member's workers' compensation claim, it relies on (a) the same employer-controlled data, available only on request and on terms the employer sets; (b) anecdote; or (c) the only independent data source presently available, which is the union's own sick-leave records.

Inventor's observation. Sick-leave records are the only hard, independent signal of workplace harm presently available to most education labor organizations — and they capture only the most severe cases, the ones where injury or trauma forced absence. The actual rate of workplace incidents is, by definition, higher than sick-leave data alone can reveal. Without a parallel reporting infrastructure owned by the labor organization, everything between the floor of sick leave and the ceiling of catastrophic events is statistically invisible.

The invention disclosed herein addresses this asymmetry directly.

5. Problem statement

The invention solves the following technical and structural problems, each of which is unaddressed or only partially addressed by prior art:

Data ownership asymmetry. Existing safety-reporting platforms are architected as employer-side systems. The employee (and the employee's labor organization) is a data subject, not a data owner. The data architecture itself encodes a conflict of interest.

Underreporting due to absent infrastructure. When the only reporting pathway is owned by the employer, workers in vulnerable categories (paraprofessionals, special-education instructional assistants, non-certificated staff) report at rates well below their actual incident exposure. The result is systematic underestimation of workplace risk.

Inability to correlate cross-source signals. Sick-leave data, incident reports, and student-behavior records exist in three different systems, controlled by three different parties, under three different regulatory regimes (HIPAA, district policy, FERPA). No prior system correlates these signals at building or district aggregate level for use by the union.

Role-appropriate visibility. An incident is simultaneously relevant to the affected staff member, that member's building representative, and the local-union president, but each of those roles needs a different view: the member sees status updates on their own report; the building rep sees building-aggregate patterns; the local president sees district-wide OSHA-threshold proximity. Prior systems either expose everything to everyone or restrict everything to administrators.

Causation attribution at the bargaining table. When incident counts rise, the employer commonly attributes the rise to 'better reporting infrastructure' rather than worsening conditions. The union has no tool that combines first-party incident data with the published epidemiological and labor-economics literature to rebut, qualify, or contextualize that argument.

Privacy-preserving student pattern tracking. A substantial fraction of K–12 workplace incidents involve students. FERPA prohibits identifying student-level information in most disclosures. But anonymized pattern data — 'Student A, Room 12, fourth bite incident in 30 days' — is exactly what staff need to advocate for BIP review, additional staffing, or manifestation-determination meetings. Prior art does not offer this primitive.

Automatic policy mapping. Each incident triggers a specific set of regulatory and contractual responses (workers' comp filing, OSHA Form 300 entry, behavior-plan review, trespass notice, threat-assessment convening). These mappings are knowable from incident attributes but in practice depend on a small number of experienced reps and presidents remembering them. Prior systems do not encode the policy map.

6. Summary of the invention

The invention is a software system, comprising a server-side data store, client-side applications, and a defined data architecture, configured such that:

- bullet the system of record is owned and operated by a labor organization (a teachers' union or education association) rather than the employer;
- bullet an affected worker (the "member") submits an incident report via a mobile or web client, with the report routed in real time to that member's building representative and the local union president;
- bullet the system maintains three distinct user roles — member, building rep, local president — with a cascading-visibility model in which each role sees the same incident framed against a different analytical context;
- bullet an analytics layer correlates aggregate sick-leave totals with incident counts at building level, producing a defensible 'ground truth' estimate of true workplace-harm rates compared to the floor visible from sick leave alone;
- bullet an anonymized student-reference system permits pattern tracking across staff and settings without disclosing student-identifying information under FERPA;
- bullet an automatic policy-suggestion engine maps incident attributes (type, severity, tags, injury qualifier) to applicable response policies, presenting suggestions to authorized roles at the moment of incident review;
- bullet a causation-analysis module overlays third-party epidemiological research onto first-party incident trends, distinguishing increases attributable to better reporting infrastructure from increases attributable to worsening conditions;
- bullet a multi-tier notification engine routes alerts by role and importance, with members receiving status updates on their own reports, building reps receiving building-level pattern alerts, and local presidents receiving district-wide threshold notifications including approaching OSHA-reportable thresholds.

Together, these features constitute a system whose primary novelty is not any single user-facing feature but the **architectural inversion** of incident reporting from an employer-owned to a labor-organization-owned paradigm, combined with the specific analytical and routing primitives needed to make that inversion operationally useful.

7. Detailed description

7.1 Data architecture

The system maintains a primary incident store containing records with the following representative schema: a globally unique identifier; a timestamp; the reporting member's identifier and union-role descriptor; the building identifier; the incident type drawn from an enumerated set (physical assault, verbal threat or harassment, weapon present, facilities hazard, other); a severity grade (low / medium / high / critical); an injury qualifier (physical / mental / both / none); a free-text description; an optional anonymized student reference; an optional set of photo references; a multi-valued tag set; a needs-immediate-assistance flag; a 911-emergency flag; a status enumerated value (new / acknowledged / in progress / resolved / escalated); a thread of role-typed reply messages; an optional set of assigned policy identifiers; and a post-incident sick-leave-days field used for ground-truth correlation.

The data store is logically and contractually owned by the labor organization. Employer access, if any, is governed by data-sharing agreements that the labor organization controls and may withdraw. This architectural property is the principal differentiator from prior art.

7.2 Three-role information model

The system defines three union-membership roles with cascading visibility:

Member. A unit-of-bargaining employee (teacher, instructional assistant, counselor, custodian, etc.). A member sees only their own reports and the status, replies, and policy assignments thereon.

Building representative. A member with elected or appointed authority for one school building. A building rep sees all incidents in their building, aggregate analytics for that building (top tags, top reporters, location-severity heat map, sick-leave correlation), and may reply to incidents, change status, and assign policies.

Local president. The executive of the local-union affiliate (typically covering a school district or county). A local president sees district-wide incidents, geographic mapping across all buildings, threshold-proximity alerts (e.g., approaching OSHA-reportable thresholds), causation-analysis overlays, and aggregate sick-leave correlation across buildings.

7.3 Real-time fan-out routing

On submission, an incident record is propagated to (a) the reporter (for confirmation and tracking), (b) the building representative for the reporter's building (selected by a building-to-rep mapping table), and (c) the local president of the labor organization (selected by an organization-to-president mapping table). For incidents flagged 911-emergency, a higher-importance notification class is generated and may also be routed to designated district safety officers if such routing is enabled by inter-party agreement.

7.4 Notification importance and role gating

Notifications are classified by importance (info / high / critical) and by type (new report, emergency, acknowledged, in progress, resolved, reply, pattern, threshold, status). Each role's notification feed is filtered by both. A member sees notifications about their own reports (typically info and high). A building rep sees building-scope notifications (typically high and critical). A local president sees district-scope notifications, including synthetic notifications generated by the analytics layer (e.g., 'Lincoln Elementary approaching OSHA threshold').

7.5 Anonymized student reference system

When an incident involves a student, the reporter may optionally select an anonymized student-reference code from a set scoped to the reporter's building. The reference resolves to a record containing a per-school anonymized identifier (e.g., 'Student A'), the student's setting (classroom or wing), grade level, existing support indicators (IEP, BIP, 504, or none), and a last-BIP-review marker. No personally identifying information about the student is stored or rendered in the incident record. The reference enables cross-staff pattern detection — for example, that three different staff members were affected by incidents tied to the same anonymized student in thirty days — while complying with FERPA's prohibitions on disclosure of personally identifiable education records.

7.6 Automatic policy-suggestion engine

Each incident is evaluated by a rule-based engine against a policy catalogue (workers' compensation filing, OSHA 300 log entry, BIP review, IEP review, FBA request, additional-staffing request, restraint documentation, trespass notice, parent conference, code-of-conduct review, threat assessment, OSHA facility complaint, manifestation determination). The engine emits a ranked list of suggested policy assignments based on incident attributes (severity, type, injury qualifier, tag set, repeat indicators). Authorized roles (building rep or local president) may assign or unassign policies, creating a tracked compliance and follow-through record for the incident.

7.7 Sick-leave correlation analytics

The analytics layer maintains a per-incident field recording the number of sick-leave days taken by the reporter in a defined window following the incident (default fourteen days). Aggregated to building or district level — never to the individual, to preserve HIPAA-adjacent privacy norms — this produces three metrics: total post-incident sick days, the percentage of incidents triggering any sick leave, and average sick-leave days per incident. These metrics constitute the system's 'hard signal' validating the underlying incident rate against an independently recorded operational source.

7.8 Causation-analysis overlay

A research-bibliography module ingests citations from published research on educator workplace violence and surfaces explanatory factors alongside the system's observed trend data. The current implementation overlays five factor categories: persistent post-pandemic dysregulation (APA 2024, CDC YRBS); special-education staffing shortage (NCES, UFT, EdResearch for Action); the spring spiral (trauma-informed K–12 research, Resilient Futures); reporting-infrastructure effect (NCES SSOCS trend data); and repeat-student compounding (Reddy et al., 2024). Each factor is linked to the corresponding first-party data signal in the system, producing a defensible attribution story rather than an unqualified count.

7.9 Geographic incident visualization

For local presidents only, a geographic visualization renders each building's incident count and maximum severity on a stylized state outline (Massachusetts in the prototype implementation). Buildings appear as colored markers sized by incident volume and colored by maximum severity, with halo-glow rendering to indicate concentration. Role restriction is deliberate: building-level geographic detail exposed to members or building reps creates social-pressure risks within buildings that are not appropriate to a safety-reporting workflow.

8. Claims preview — the novel features

The following are claim-style summaries of features the inventors believe to be novel and non-obvious in light of prior art. They are written in plain English rather than patent-claim form; conversion to enforceable claim language is the task of qualified counsel. They are non-exhaustive and presented in approximate order of inventive significance.

CLAIM 01

Union-owned incident-reporting data architecture

A safety-incident reporting system in which the system of record, the access controls, and the analytical surfaces are owned and operated by a labor organization representing the affected workforce, with employer access (if any) governed by data-sharing terms set by the labor organization.

CLAIM 02

Cascading three-tier role visibility on a single incident

A method for rendering a single incident record across three union-role views — member, building representative, local president — such that each role is shown the incident accompanied by analytical context appropriate to that role's scope of authority and responsibility.

CLAIM 03

Sick-leave-to-incident aggregate correlation as ground truth

A method for correlating, at building or district aggregate level, the number of sick-leave days taken by reporters in a defined post-incident window against the count and severity of incident reports filed in the same period, for the purpose of estimating true workplace-harm rates from an independently validated operational signal.

CLAIM 04

Anonymized student-reference primitive for FERPA-compliant pattern tracking

A data model in which student-involved incidents are linked to per-school anonymized student-reference codes, such that cross-staff and cross-location patterns can be detected and surfaced without storing or rendering personally identifying information about the student.

CLAIM 05**Automatic incident-to-policy suggestion engine**

A rule-based system that maps incident attributes (severity, type, injury qualifier, tag set, repeat indicators) to a ranked list of applicable response policies drawn from a configurable catalogue spanning regulatory (OSHA Form 300, Workers' Compensation), educational (IEP, BIP, FBA, manifestation determination), and contractual (trespass notice, parent conference) response classes.

CLAIM 06**Causation-analysis overlay combining first-party data with published research**

A method for presenting trend data alongside ranked third-party epidemiological and labor-economics research, with each research finding linked to a corresponding first-party data signal, producing attributable explanations rather than unqualified counts.

CLAIM 07**Role- and importance-gated synthetic notification generation**

A notification system that generates synthetic alerts — including pattern detection (e.g., recurring location), threshold proximity (e.g., approaching OSHA-reportable counts), and demographic concentration (e.g., share of incidents involving a specific staff role) — and routes them by role and importance class.

CLAIM 08**Geographic incident visualization with role-restricted disclosure**

A geographic rendering of incident counts and severities scoped to executive union roles only, deliberately withheld from non-executive roles to prevent intra-building social pressure dynamics.

CLAIM 09**Two-way real-time threading with role identity preservation**

An incident-reply thread in which each message is tagged with the speaker's union role at the time of posting, such that the historical record reflects who spoke in what capacity, supporting subsequent grievance, workers'-compensation, or bargaining use.

CLAIM 10**Bargaining-intent telemetry surface**

An analytics surface specifically configured to render patterns relevant to collective bargaining (proportion of incidents involving a low-paid worker classification, staffing-ratio inferences, repeated-location concentration) rather than to employer disciplinary or compliance use.

Note on patentability. Software claims face elevated scrutiny under *Alice Corp. v. CLS Bank International*, 573 U.S. 208 (2014), and *Mayo Collaborative Services v. Prometheus Laboratories*, 566 U.S. 66 (2012). Counsel should evaluate each of the above claims under the two-step *Alice/Mayo* framework: (i) is the claim directed to a patent-ineligible concept such as an abstract idea? (ii) if so, does the claim recite 'significantly more' than the abstract idea itself? Claims rooted in specific technical architecture (Claim 01), specific data-model primitives (Claim 04), or specific cross-source correlation methods (Claim 03) are likely stronger candidates than claims directed primarily to information presentation.

9. Prior art comparison

The inventors have identified the following categories of prior art. A formal prior-art search by qualified counsel or a registered search firm is recommended before any filing.

9.1 District-side incident management platforms

Examples: Frontline Education (incident-management module), PowerSchool Behavior, Educator's Handbook, district-developed proprietary forms. **Differentiation:** these systems are employer-owned by construction. They do not offer union-role views, union-owned data, sick-leave correlation, FERPA-compliant cross-staff pattern tracking, or bargaining-oriented analytics.

9.2 General workplace incident reporting platforms

Examples: Origami Risk, SafetyCulture (iAuditor), Intellex, RiskCloud. **Differentiation:** general-purpose; not configured for K–12; not configured for labor-organization ownership; do not address FERPA constraints; do not provide policy mapping to educator-specific responses (BIP, IEP, manifestation determination); no sick-leave correlation.

9.3 Union organizing and member-engagement tools

Examples: UnionTrack, BroadStripes, Unionware, OurUnion. **Differentiation:** these are organizing and CRM tools (membership records, dues, campaigns, internal communications). They do not provide incident reporting, safety analytics, or bargaining intelligence. Combining a union CRM with the invention disclosed here is not anticipated by prior art.

9.4 Workers' compensation and OSHA filing systems

Examples: state-administered Workers' Comp portals, OSHA Form 300 templates, state Department-of-Labor reporting tools. **Differentiation:** these are downstream filing endpoints, not real-time incident-management systems. The invention here may interoperate with them (suggesting and tracking filings) but does not duplicate them.

9.5 Anonymous tip lines and student-safety reporting

Examples: SafeUT, P3 Campus, Safe2Tell, Sandy Hook Promise 'Say Something'. **Differentiation:** these are student-to-administration channels, often anonymous, designed for threat reporting and student welfare. They are not staff-to-union channels; they do not preserve the speaker's identity for workers'-compensation or grievance use; they are operationally owned by third-party security vendors or districts.

9.6 General-purpose analytics dashboards

Examples: Tableau, Looker, Power BI used over district incident data. **Differentiation:** general-purpose tooling. Does not provide the specific primitives disclosed here (anonymized student reference, policy suggestion, causation overlay, sick-leave correlation, role-gated views). The combination of those primitives in a single integrated workflow is the inventive contribution.

10. Embodiments and implementation

10.1 Reference implementation

A working software prototype implementing the invention has been reduced to practice. The current prototype is a single-page web application written in JavaScript using the React user-interface library (version 18.3), the Recharts data-visualization library (version 2.12), and Lucide React iconography (version 0.383). The codebase comprises approximately 3,700 lines of JavaScript and JSX organized as a single application file. Persistence in the prototype is via the browser's localStorage facility; production implementations would substitute a server-side data store with appropriate encryption, audit logging, and access controls.

10.2 Anticipated alternative embodiments

The invention is not limited to the reference implementation. Counsel should consider claim language that captures the following alternative embodiments:

- bullet Native iOS and Android mobile applications with offline-capable incident capture and queued synchronization upon reconnection;
- bullet Progressive web application (PWA) implementations with service-worker-mediated offline support;
- bullet Server-side implementations using any back-end framework (Node.js, Python/Django, Ruby/Rails, Go) and any compliant data store (PostgreSQL, MySQL, document stores);
- bullet Implementations integrating with Student Information Systems (Aspen, PowerSchool, Skyward, Infinite Campus, etc.) via published APIs to auto-populate anonymized student references;
- bullet Implementations integrating with payroll or HRIS systems to source sick-leave correlation data;
- bullet Implementations extending the role hierarchy beyond three tiers, e.g., adding state-affiliate or national-affiliate views over multiple locals;
- bullet Implementations across non-education labor sectors (healthcare nursing unions, public-safety unions, transit unions) with the policy-catalogue swapped for the sector's applicable rules;
- bullet Implementations within district-side software (a dual-SKU offering: union version and employer version) where the data-architectural inversion is preserved by contract rather than by deployment.

11. Trademark considerations

The inventors intend to use the mark **SAFEPOINT** in commerce in connection with the goods and services described herein. The mark is presently in use within a pre-launch prototype context.

Suggested classes for filing

IC 9	Downloadable mobile application software for safety incident reporting; downloadable software for workplace-safety analytics.
IC 42	Software-as-a-service (SaaS) services featuring software for safety incident reporting, real-time alerting, and analytics for labor organizations.
IC 45	Provision of information relating to workplace safety; data services for labor unions and labor organizations.

Mark variants to consider protecting

(i) The word mark SAFEPOINT in standard characters; (ii) the visual logo mark (presently a rust-colored square with a centered white inset block); (iii) the combined mark; (iv) selected slogans and taglines used commercially (e.g., 'Reporting that stays with the union' or similar, subject to final selection).

Pre-filing search

A prudent next step is a USPTO Trademark Electronic Search System (TESS) clearance search and a common-law search before any intent-to-use filing under Section 1(b) of the Lanham Act. The mark 'SafePoint' is in use by other parties in unrelated classes (consulting, publishing, security); coexistence in IC 9 / 42 / 45 for the present goods and services appears plausible but should be confirmed by counsel.

12. Copyright considerations

Copyright in original works of authorship attaches automatically upon fixation in a tangible medium under 17 U.S.C. § 102(a). The following works are subject to copyright protection in the inventors' favor:

- bullet the source code of the reference implementation, as fixed in version-controlled files;
- bullet the user-interface design as expressed in screen layouts, typography choices, color systems, and visual compositions;
- bullet the textual content of the application, including all microcopy, error messages, field labels, help text, and explanatory commentary;
- bullet the research-citation commentary in the causation-analysis module, to the extent it constitutes original expression rather than mere recitation of facts;
- bullet the documentation and explanatory writings including this Invention Disclosure Document;
- bullet any logos, marks, and visual assets to the extent they constitute pictorial or graphic works.

Registration

While registration is not required for copyright to exist, registration with the U.S. Copyright Office is required before bringing an infringement suit for U.S. works (17 U.S.C. § 411(a)) and confers eligibility for statutory damages and attorneys' fees (17 U.S.C. § 412). The inventors should register the source code and the user-interface design with the Copyright Office; software is filed on Form TX with a deposit of identifying portions of source code (typically the first and last twenty-five pages with trade-secret material blocked).

Open-source licensing decision

The current reference implementation is distributed in source form under the MIT License. This is a deliberate choice intended to maximize adoption among labor organizations and academic researchers; it does not preclude separate commercial licensing of derivative works or hosted-service operation. Counsel should evaluate whether the MIT License meaningfully reduces patent enforceability or trade-secret protectability and recommend appropriate licensing terms going forward.

13. Trade secret considerations

Certain components of the invention may be more durably protected as trade secrets than as patents, including:

- bullet the specific weighting and tuning of the policy-suggestion engine's rules;
- bullet the empirical thresholds and decay functions used by the synthetic-notification engine (e.g., the specific incident count or window that triggers an 'approaching OSHA threshold' alert);
- bullet the bibliography curation and the mapping of research findings to first-party data signals in the causation-analysis module;
- bullet any forthcoming proprietary integrations with payroll or HRIS systems and the specific contractual terms governing sick-leave data sharing.

Trade-secret protection requires that the holder take reasonable measures to maintain secrecy (18 U.S.C. § 1839(3)(A)). Recommended measures include: written non-disclosure agreements with any party (employees, contractors, pilot-program partners) exposed to the protected material; access controls limiting trade-secret material to need-to-know personnel; written employee agreements containing intellectual-property assignment and confidentiality covenants; and explicit confidentiality marking on all materials.

14. Recommended next steps

01. Engage IP counsel. Retain a registered U.S. patent attorney or agent with experience in software-related inventions and a working familiarity with the labor-sector context. This Invention Disclosure Document is intended to serve as the engagement-meeting briefing material.

02. File a provisional patent application. Under 35 U.S.C. § 111(b), a provisional application establishes a priority date for up to twelve months at significantly reduced cost compared to a non-provisional. The provisional should incorporate this disclosure essentially verbatim, with figures added by counsel.

03. Conduct a formal prior-art search. Before incurring non-provisional filing costs, engage a registered search firm or patent attorney to perform a formal prior-art search using USPTO, EPO, and WIPO databases. Adjust claim scope based on search results.

04. File a trademark intent-to-use application. File an ITU application under Lanham Act § 1(b) for the word mark SAFEPOINT in IC 9, 42, and 45 immediately after a TESS clearance search. Statement of Use to follow first commercial offering.

05. Register key copyrighted works. Within three months of first publication, register the source code (Form TX) and the user-interface design with the U.S. Copyright Office to preserve statutory-damages eligibility.

06. Implement trade-secret hygiene. Adopt a written confidential-information policy. Execute NDAs with all contractors, advisors, and pilot-program participants. Mark confidential documents (including this one) accordingly. Restrict access to trade-secret-designated material.

07. Defer non-essential public disclosure. Avoid public demonstrations, conference presentations, press coverage, and commercial offers for sale until at least the provisional application is on file. The one-year U.S. grace period is not available in most foreign jurisdictions.

08. Document inventor and joint-inventor contributions. If any other party may have contributed to the conception of any claim, document the contribution contemporaneously and consult counsel on joint inventorship before filing.

15. Inventor declaration and signatures

We, the undersigned, each hereby declare that we are joint inventors of the subject matter disclosed in this Invention Disclosure Document; that the subject matter was first conceived by us on the date stated in Section 2 of this document; that each of us contributed to the conception of at least one claim of the invention as described in the contribution statements in Section 2; that to the best of our knowledge no other party has independently conceived the same subject matter; that we have not knowingly publicly disclosed, commercially offered for sale, or placed into public use the subject matter prior to the date of this disclosure other than as expressly noted herein; and that the contents of this disclosure are accurate and complete to the best of our knowledge and belief.

Joint inventor 1

Signature:

Printed name:

Sonny Steele

Date:

Joint inventor 2

Signature:

Printed name:

Nick Sireci

Date:

Witness

Signature:

Printed name:

Date:

Witness instructions. The witness should be a person who has read and understood the substance of this disclosure but is not a joint inventor. The witness's role is evidentiary: to establish, if later contested, that the

disclosure existed in its present form on or before the date of signature. A witness who is technically competent to understand the disclosure is preferred but not required.

Appendix A. Code repository reference

The reference implementation of the invention is maintained as a source-code repository. As of the date of this disclosure, the principal application file is `src/App.jsx` within the repository, comprising approximately 3,700 lines of JavaScript and JSX. Build configuration is via Vite. The repository includes a README, license (MIT), and the structured build files necessary for a third party to reproduce the working prototype with `npm install && npm run dev`.

Counsel and any registered prior-art searcher should request the repository snapshot corresponding to the conception date stated in Section 2 as evidence of reduction to practice.

Major source modules in the reference implementation

- bullet `App` — application root and routing
- bullet `ReportForm` — member-side incident capture (Section 7 architecture)
- bullet `MemberHome`, `RepDashboard`, `PresidentDashboard` — the three role views
- bullet `NotificationCenter` — role- and importance-gated alert feed
- bullet `GeoHeatMap` — geographic visualization (president-only)
- bullet `TrendCausationPanel` — causation overlay
- bullet `StudentPatterns` — FERPA-compliant student pattern primitive
- bullet `SickLeaveCallout`, `SickTimeByBuildingList` — sick-leave correlation surfaces
- bullet `PolicyAssign`, `suggestPolicies` — policy-suggestion engine
- bullet `Term`, `TermText` — in-context acronym glossary

Appendix B. Research bibliography

The following published research informs the causation-analysis module described in Section 7.8 and supports the field-of-invention statement in Section 3.

[01] American Psychological Association & National Education Association (2025). *Violence Against Educators: APA-NEA Policy Brief*. (Issued November 2025.) Available at nea.org.

[02] Reddy, L. A., Perry, A. H., Martinez, A., et al. (2024). 'Student Violence Against Paraprofessionals in Schools: A Social-Ecological Analysis of Safety and Well-Being.' *Behavioral Sciences* 14(12), 1181.

[03] Espelage, D., et al. (2024). APA Task Force on Violence Against Educators report. *American Psychologist*.

[04] National Center for Education Statistics (2024). *School Survey on Crime and Safety (SSOCS) 2021-22*. U.S. Department of Education.

[05] National Center for Education Statistics & Bureau of Justice Statistics (2024). *Indicators of School Crime and Safety: 2023*.

[06] Federal Bureau of Investigation (2025). *Crime in Schools, 2020–2024*. Special report, August 2025.

[07] United Federation of Teachers (2024). Special-education vacancy report, November 2024.

[08] EdResearch for Action (2024). *Brief #31: Addressing Special Education Staffing Shortages*.

[09] Resilient Futures (2026). 'The Spring Spiral: Why Student Behavior Escalates This Time of Year.' Trauma-informed K–12 research brief.

[10] Massachusetts Department of Labor Standards. *Workplace Violence Prevention Program Elements for K–12 Schools*.

[11] Occupational Safety and Health Act, 29 U.S.C. § 654(a)(1) (General Duty Clause).

[12] Family Educational Rights and Privacy Act, 20 U.S.C. § 1232g; 34 C.F.R. Part 99.

[13] Individuals with Disabilities Education Act, 20 U.S.C. § 1400 et seq.

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